

Chan Ding Hao

+65 8202 3989 | chandinghao@yahoo.com | [Github](#) | [LinkedIn](#) | [Portfolio Website](#) | [Medium](#)

Education

Singapore Management University - Bachelor of Science in Economics

Aug 2022 - July 2026 (Expected)

- **Relevant Coursework:** Big Data Architecture, Machine Learning & Applications, Algorithms and Programming, Visual Analytics for Business Intelligence, Business Data Management, Computational Thinking & Programming, Data Analytics with R

Work Experience

ByteDance - Payment Risk Data Analytics Intern

Jan 2026 - July 2026

- Built end-to-end risk scoring system for TikTok Live onboarding that evaluates account features (IP geolocation, device fingerprinting, behavioral signals) to detect and action against users creating spam accounts
- Engineered production ETL pipelines in SparkSQL/Hive to power user bot detection models, processing 1,000,000 rows of transaction data daily for ByteDance's anti bot detection system
- Built operational dashboards to monitor fraud detection rule performance and track payment risk KPIs, enabling risk analysts to identify rule degradation and optimize model effectiveness in real-time

Razer - Data Analyst Intern (Analytics Engineering)

July 2025 - Dec 2025

- Built dashboards on Apache Superset to track key metrics (Daily CTR, Daily Revenue etc) for Razer recommendation engine using SQL, reducing reporting time by 25%.
- Built transformation models (SQL) within ELT pipelines using dbt DAGs to deliver business intelligence solutions for internal stakeholders served through Apache Superset, increasing dashboard load speed by 50%.
- Developed **deduplication logic** for user software data through **aws wrangler**, **SQL** and **pandas** in **AWS SageMaker AI** notebooks, identifying data quality issues and providing implementation specifications to data engineering.

ONE (Ocean Network Express) - Data Analyst Intern

June 2024 - Dec 2024

- Optimized truck routes using a **CI/CD pipeline on GCP**, leveraging **Python** for data transformation and analysis on **VertexAI** and **SQL** on **BigQuery**. **Improved processing speed by 10%**, enabling faster data-driven decisions and improving real-time logistics efficiency.
- Discovered important aspects related to data (e.g. Hazmat constraints), while validating discoveries with various stakeholders, **reducing distance by 5%, and saving over \$500,000 USD annually with scalability potential**.
- Analyzed **SQL** queries on **BigQuery** to trace undocumented data sources, enabling accurate data analysis for route optimisation.

Projects

Crypto Market Data Ingestion (In Progress)

Jan 2026 - April 2026 (Expected)

- Built end-to-end crypto market intelligence pipeline on AWS (EventBridge → Lambda → RDS PostgreSQL) ingesting daily BTC/USD OHLC snapshots from Kraken REST API, enabling exchange-focused business metrics analysis
- Designed and deployed Grafana dashboards tracking 5 key exchange performance metrics — trading volume, VWAP vs. close, volume-volatility correlation, 30-day volatility regime, and drawdown — to surface actionable revenue and liquidity insights

SPY Analytics and Forecasting - [Github](#) - [Medium \(1\)](#), [\(2\)](#), [\(3\)](#)

Aug 2024 - Dec 2024

- Conducted extensive **data wrangling, analysis, and visualisation** of SPY's historical and inflation-adjusted returns alongside Federal Reserve inflation data to highlight long-term investment advantages and performance stability.
- Applied **time-series statistical techniques**, such as **stationarity testing (ADF)** and **differencing**, to preprocess SPY data for robust time series forecasting.
- Constructed a weighted ensemble model (**LSTM -PyTorch**, **Prophet**, and **Monte Carlo simulations**) for **time series forecasting**. Predicted the log returns of SPY in the next 20 years with confidence intervals.

Skills and Certifications

Technical Skills: Python; PySpark; SQL (HiveQL, SparkSQL, MySQL, PostgreSQL); R; dbt; Bash/Linux; Git; Docker

Libraries & Frameworks: NumPy; Pandas; PyTorch; Scikit-learn; XGBoost; Plotly; Boto3; AWS Wrangler; Google Cloud SDK; Streamlit

Cloud & Platforms: GCP (BigQuery, Vertex AI); AWS (S3, Redshift, RDS, Lambda, SageMaker, Bedrock); Pinecone

BI & Visualization: Apache Superset; Grafana; Tableau

Certifications & Training [Google Data Analytics Professional Certificate](#), [HackerRank SQL \(Advanced\)](#)